

# File permissions in Linux

## Project description

In this scenario, I was tasked with updating the file permissions on `researcher2`'s account, so that the principle of least privilege was preserved. Some files were configured such that owners, groups, and other users had more access than necessary or wise. My job was to rectify these issues using the `chmod` command, to close security gaps and protect sensitive information from unauthorized access.

## Check file and directory details

I first displayed the permissions for each file and directory, including hidden files, using the `ls -al` command:

```
researcher2@762db49f0d78:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:26 ..
-rw--w---- 1 researcher2 research_team  46 Jun 27 17:11 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 27 17:11 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 27 17:11 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 27 17:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_t.txt
researcher2@762db49f0d78:~/projects$
```

## Describe the permissions string

An example permissions string is the one for `project_k.txt`: `-rw-rw-rw-`. The string indicates that the user (owner), group (to which the owner belongs), and other users (everyone else) are all allowed to read and write to, but not execute, the file. The following chart breaks down each character in the string:

| Character | Meaning                                                         |
|-----------|-----------------------------------------------------------------|
| -         | File type: - for a file, d for a directory                      |
| r         | User read: r for read permissions, - for no read permissions    |
| w         | User write: w for write permissions, - for no write permissions |

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| -              | User execute: <code>x</code> for execute permissions, <code>-</code> for no execute permissions  |
| <code>r</code> | Group read: <code>r</code> for read permissions, <code>-</code> for no read permissions          |
| <code>w</code> | Group write: <code>w</code> for write permissions, <code>-</code> for no write permissions       |
| -              | Group execute: <code>x</code> for execute permissions, <code>-</code> for no execute permissions |
| <code>r</code> | Other read: <code>r</code> for read permissions, <code>-</code> for no read permissions          |
| <code>w</code> | Other write: <code>w</code> for write permissions, <code>-</code> for no write permissions       |
| -              | Other execute: <code>x</code> for execute permissions, <code>-</code> for no execute permissions |

## Change file permissions

Then I removed write permissions for other users from `project_k.txt` and read permissions for the group from `project_m.txt`. I used `chmod o-w project_k.txt` to remove (-) write permissions (`w`) from other (`o`) users. I used `chmod g-r project_m.txt` to remove (-) read permissions (`r`) from the group (`g`).

```
researcher2@762db49f0d78:~/projects$ chmod o-w project_k.txt
researcher2@762db49f0d78:~/projects$ chmod g-r project_m.txt
researcher2@762db49f0d78:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:26 ..
-rw--w---- 1 researcher2 research_team  46 Jun 27 17:11 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 27 17:11 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 27 17:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_t.txt
researcher2@762db49f0d78:~/projects$
```

## Change file permissions on a hidden file

Next, I changed the file permissions on the hidden file `.project_x.txt` so that the user and group can read but not write or execute. I used `chmod u=r,g=r .project_x.txt` to assign (=) read permissions (`r`) to both the user (`u`) and the group (`g`), overwriting any previous permissions.

```
researcher2@762db49f0d78:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@762db49f0d78:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:26 ..
-r--r----- 1 researcher2 research_team  46 Jun 27 17:11 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 27 17:11 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 27 17:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_t.txt
researcher2@762db49f0d78:~/projects$
```

## Change directory permissions

Finally, I removed group execute permissions from the `drafts` subdirectory. I used `chmod g-x drafts/` to remove (-) execute permissions (x) from the group (g).

```
researcher2@762db49f0d78:~/projects$ chmod g-x drafts/
researcher2@762db49f0d78:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 27 17:26 ..
-r--r----- 1 researcher2 research_team  46 Jun 27 17:11 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 27 17:11 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 27 17:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 27 17:11 project_t.txt
researcher2@762db49f0d78:~/projects$
```

## Summary

In conclusion, using the `ls` and `chmod` commands, I was able to reconfigure file and directory permissions on a Linux account, in order to preserve the principle of least privilege. I examined file and directory permissions and restricted the access of certain users and groups.